



TECHNICAL NOTICE

Third-party Runtime Components

MediaPipe provenance, modifications, checksums, licensing, and update procedure.

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MediaPipe 0.10.18 ARM64 wheel

A Python wheel is a standard installable package archive with a `.whl` file extension. It is similar to a ZIP file, but follows a defined layout that lets `pip` install Python modules, native libraries, and dependency metadata without building the project from source on the target device. In this case, the wheel contains MediaPipe's Python package and its already-compiled Linux ARM64 native code, so the UNO Q does not have to compile MediaPipe itself.

The filename describes its compatibility: `cp312-cp312` means CPython 3.12 and its matching application binary interface, while `manylinux2014_aarch64` and `manylinux_2_17_aarch64` identify compatible 64-bit ARM Linux environments. The UNO Q runtime uses this tracked wheel:

```
python/worker-wheels/mediapipe-0.10.18-cp312-cp312-manylinux2014_aarch64.manylinux_2_17_aarch64.whl
```

Property	Value
Component	MediaPipe 0.10.18 for CPython 3.12, Linux ARM64
Upstream project	< https://github.com/google-ai-edge/mediapipe >
Upstream package	< https://pypi.org/project/mediapipe/0.10.18/ >
License	Apache License 2.0
PowerGlove repack SHA-256	f2617c0960eb35aaa58b76076a9ae629edbf7ec8901d5fab4b046c28d13fbe8d
Original PyPI wheel SHA-256	09cbf7dc1f9a2deaaaac687e5f982836623def4cbd3e827d95f86f42450d2dd1

Modification notice

PowerGlove Vision repackaged the upstream wheel for its headless UNO Q worker. The Python package code and compiled MediaPipe binaries were not modified. The wheel dependency metadata was changed as follows, and its `RECORD` was rebuilt:

- removed the `jax` dependency declaration;
- removed the `jaxlib` dependency declaration;
- replaced `opencv-contrib-python` with `opencv-contrib-python-headless==4.10.0.84`;
- omitted the upstream wheel's empty `mediapipe.libs` directory.

These changes avoid unnecessary JAX installation and GUI OpenCV dependencies on the UNO Q. The repacked wheel retains MediaPipe's Apache 2.0 license at `mediapipe-0.10.18.dist-info/LICENSE`. Do not substitute the upstream wheel without retesting dependency resolution, camera startup, and hand tracking.

Google Hand Landmarker model

PowerGlove Vision uses Google's float16 Hand Landmarker task bundle.

Property	Value
UNO Q runtime path	data/models/hand_landmarker.task
Official download	< https://storage.googleapis.com/mediapipe-models/hand_landmarker/hand_landmarker/float16/1/hand_landmarker.task >
SHA-256	fbc2a30080c3c557093b5ddfc334698132eb341044ccee322ccf8bcf3607cde1
Size	7,819,105 bytes

The model is not tracked in Git and is not included in the App Lab installation ZIP. On first launch, the application downloads it from Google's versioned URL into the private, persistent [data/models/](#) directory. It refuses to start the vision worker if the SHA-256 checksum differs, reports the problem on the dashboard, and retries without taking the web interface offline. Later launches reuse the verified cached model. Wi-Fi deployments preserve [data/](#), so an application update does not download the model again.

[scripts/fetch-runtime-assets.sh](#) provides an optional manual prefetch using the same URL and checksum. It writes to the same private [data/models/](#) path and is useful for development or for preparing the app before an offline session.

Google's MediaPipe repository is Apache-2.0 licensed and its official examples direct applications to this model URL. The model download does not place a separate model-specific license file in this repository. The PowerGlove Vision MIT License therefore should not be interpreted as licensing the Google model. The project avoids redistributing the model by having each installation fetch it directly from Google. Anyone who independently bundles or redistributes the model should confirm that the intended distribution complies with Google's applicable model terms.

Documentation illustration provenance

The gesture sheets under [docs/images/gestures/](#) were generated on September 3, 2026 with OpenAI's image-generation tool from project-authored prompts, then selected and arranged for the PowerGlove Vision gameplay guide. They are documentation assets, not runtime dependencies. No game screenshots, scans, box art, characters, publisher logos, or other source images were supplied to the generator.

The repository applies its MIT License to these curated project assets to the extent the project owner has rights in them. Game names and other third-party marks remain the property of their respective owners.

Updating either component

Treat a wheel or model update as a tested dependency change:

- 1 Record the official source URL, version, license, size, and SHA-256 here.
- 2 Update the pinned value in [scripts/fetch-runtime-assets.sh](#) when changing the model.
- 3 If repackaging another wheel, record every difference from upstream and retain its license files.
- 4 Build the App Lab installation ZIP and confirm it contains one wheel, no model file, and only the root [sketch/](#) application sketch.
- 5 Test first-launch download and checksum verification, camera initialization, tracking, the Learn and Debug pages, and controller output on the UNO Q before publishing the package.